

Causal Inference About the Effects of Interventions From Observational Studies in Medical Journals

Issa J. Dahabreh, MD, ScD; Kirsten Bibbins-Domingo, PhD, MD, MAS

IMPORTANCE Many medical journals, including *JAMA*, restrict the use of causal language to the reporting of randomized clinical trials. Although well-conducted randomized clinical trials remain the preferred approach for answering causal questions, methods for observational studies have advanced such that causal interpretations of the results of well-conducted observational studies may be possible when strong assumptions hold. Furthermore, observational studies may be the only practical source of information for answering some questions about the causal effects of medical or policy interventions, can support the study of interventions in populations and settings that reflect practice, and can help identify interventions for further experimental investigation. Identifying opportunities for the appropriate use of causal language when describing observational studies is important for communication in medical journals.

OBSERVATIONS A structured approach to whether and how causal language may be used when describing observational studies would enhance the communication of research goals, support the assessment of assumptions and design and analytic choices, and allow for more clear and accurate interpretation of results. Building on the extensive literature on causal inference across diverse disciplines, we suggest a framework for observational studies that aim to provide evidence about the causal effects of interventions based on 6 core questions: what is the causal question; what quantity would, if known, answer the causal question; what is the study design; what causal assumptions are being made; how can the observed data be used to answer the causal question in principle and in practice; and is a causal interpretation of the analyses tenable?

CONCLUSIONS AND RELEVANCE Adoption of the proposed framework to identify when causal interpretation is appropriate in observational studies promises to facilitate better communication between authors, reviewers, editors, and readers. Practical implementation will require cooperation between editors, authors, and reviewers to operationalize the framework and evaluate its effect on the reporting of empirical research.

JAMA. 2024;331(21):1845-1853. doi:10.1001/jama.2024.7741
Published online May 9, 2024.

← Editor's Note page 1812

+ Supplemental content

+ CME at jamacmelookup.com

Author Affiliations: CAUSALab, Harvard T.H. Chan School of Public Health, Boston, Massachusetts (Dahabreh); Department of Epidemiology, Harvard T.H. Chan School of Public Health, Boston, Massachusetts (Dahabreh); Department of Biostatistics, Harvard T.H. Chan School of Public Health, Boston, Massachusetts (Dahabreh); Smith Center for Outcomes Research in Cardiology, Beth Israel Deaconess Medical Center, Boston, Massachusetts (Dahabreh); Statistical Editor, *JAMA* (Dahabreh); Department of Medicine, University of California, San Francisco (Bibbins-Domingo); Department of Epidemiology and Biostatistics, University of California, San Francisco (Bibbins-Domingo); Editor in Chief, *JAMA* and *JAMA Network* (Bibbins-Domingo).

Corresponding Author: Issa J. Dahabreh, MD, ScD, CAUSALab, Department of Epidemiology, Harvard T.H. Chan School of Public Health, 677 Huntington Ave, Room 816C, Boston, MA 02115 (idahabreh@hsph.harvard.edu).

Many medical journals, including *JAMA*, restrict the use of causal language to describing studies in which the intervention is randomly assigned. Indeed, randomized clinical trials are widely viewed as the preferred way of answering questions about the causal effects of interventions. Yet it is not feasible to answer all such questions with trials due to limitations including cost, follow-up duration, or ethical considerations. When such limitations preclude the conduct of trials, carefully designed analyses of observational (nonexperimental) data offer an alternative source of evidence on the effects of interventions (eg, treatment strategies, policies, or changes in behavior). Furthermore, observational studies can serve as a data-driven approach for identifying interventions that merit further experimental investigation and for examining the effects of interventions in populations and settings that reflect practice.

The potential of observational studies to contribute evidence about the causal effects of interventions is actively being exam-

ined across medicine, epidemiology, biostatistics, economics, and other social sciences. In this Special Communication, we examine a framework that might be used by medical journals as they move away from the current approach prohibiting the use of any causal language for observational studies and toward a more comprehensive approach for causal inference that reflects a synthesis of extensive prior work spanning multiple, diverse disciplines. We undertake this examination now for 3 main reasons. First, decision-makers are increasingly seeking timely answers to complex research questions about the effects of interventions that are challenging or impossible to address with randomized trials. For example, questions about long-term or rare effects of treatment, heterogeneity of treatment effects, or the effects of health care policies can be difficult to answer by relying exclusively on trials. Second, there has been wide dissemination of frameworks for posing causal questions and elaborating the assumptions needed to answer them.¹⁻⁴¹ These frameworks have supported the refinement of existing methods and the

development of new methods that promise to deliver results that have a causal interpretation, provided strong assumptions are met.⁴²⁻¹³⁸ Third, observational data from multiple sources (eg, registries, health care claims, electronic health records) are increasingly available for research purposes. Analyses from different sources can facilitate the evaluation of robustness by using data with different measurement characteristics from populations that may have different underlying causal structures.

In what follows, we first lay out the challenges inherent in drawing causal inferences about the effects of interventions from observational studies. We then discuss limitations of the current approach to determining the appropriateness of causal language for observational studies. Finally, we propose an alternative framework for causal inference in medical and health policy research and examine its implications for authors, reviewers, editors, and readers of clinical journals.

The Challenge in Drawing Causal Inferences From Observational Studies

Increasing use of observational studies to address questions about the causal effects of interventions poses a challenge to journals that primarily serve clinical audiences. These observational studies depend more heavily on causal and statistical modeling assumptions compared with large, well-conducted randomized trials. Therefore, all other study aspects being equal, drawing causal inferences from observational studies is inherently more speculative. But, as noted earlier, all other study aspects are often not equal. Randomized trials cannot address all causal questions of importance in medicine and health policy and may have limited generalizability; thus, investigators may need to use observational studies as a source of evidence to address causal questions. The challenge, then, is to balance the importance of addressing the causal questions for which observational studies are needed with caution regarding the reliance on strong assumptions to support causal conclusions.

When researchers are confronted with this challenge, one response is to retreat from causal goals and pursue purely descriptive or predictive goals for observational studies. This approach often amounts to applying a randomization-centered criterion for determining whether causal language is allowed, resulting in exclusively associational language for any investigation using observational data. With this approach, a single study design element essentially dictates the language that can be used to describe goals, methods, and interpretations. For example, current Instructions for Authors in *JAMA* and the *JAMA Network* journals state that “[c]ausal language (including use of terms such as effect and efficacy) should be used only for randomized clinical trials. For all other study designs..., methods and results should be described in terms of association or correlation and should avoid cause-and-effect wording.” This recommendation is also included in the *AMA Manual of Style*.¹³⁹ Nevertheless, rare ad hoc exceptions have been made by *JAMA* and *JAMA Network* journals in allowing causal interpretations for observational analyses in which necessary assumptions were articulated and deemed plausible.^{140,141} Furthermore, articles in the *JAMA Guide to Statistics and Methods* series have discussed various causal inference methods.¹⁴²⁻¹⁵¹

Limitations of the Randomization-Centered Criterion for Determining the Appropriateness of Causal Language and Interpretation

The use of a binary, randomization-centered criterion for allowing the use of causal language or interpretation is not problematic when applied to large, well-conducted randomized trials with near-perfect adherence to the study protocol and limited missing outcomes, wherein a causal interpretation is warranted. However, for many other studies, the approach based on this criterion is inadequate and does not accommodate precise descriptions of goals, research questions, methods, assumptions, and interpretations, and can result in lack of clarity during interactions among authors, editors, reviewers, and readers. The prohibition impedes the presentation and critique of study methods and risks misinterpretation of results both by allowing inappropriately drawn implicit causal inferences and by obscuring appropriate causal conclusions.

Prohibiting causal language when describing observational studies does not allow authors to communicate their research goals clearly and fully.¹⁵²⁻¹⁵⁴ Causal goals require causal assumptions (eg, the assumption of no uncontrolled confounding). These assumptions are almost never possible to verify with the data alone, and their plausibility can best be assessed within an explicit causal framework. Without causal language, the description and critique of research methods becomes challenging because the connection between ends (causal goals) and means (research methods) is obscured.¹⁵⁴ Furthermore, when causal goals, assumptions, and methods cannot be explicitly discussed, assessing the choice of study design and analytic approaches and interpreting results become difficult, if not impossible. In fact, avoidance of causal language precludes effective criticism grounded in causal considerations. For example, if a manuscript purports to present only descriptive or predictive associations between some exposure (or treatment) and outcomes, there is little room for discussing confounding in the sense of comparability between intervention groups.^{153,154} Yet such discussion is often necessary to uncover the reported study’s limitations if a causal interpretation is under consideration. In other words, restricting causal discourse is undesirable because authors and readers often hope that the estimated associations have a tenable causal interpretation and are interested to know when and why such interpretation may not be valid.

In addition, using a single study design element (randomization) as the sole criterion of whether causal conclusions can be drawn risks giving the impression of complacency about potential weaknesses that can affect both randomized trials and observational studies. Editors, reviewers, and readers would not draw causal conclusions based on simple between-treatment-group comparisons from a randomized trial with poor data collection practices, differential outcome ascertainment, or a high dropout rate, but these issues are not given the same weight as (lack of) randomization when the current approach to the use of causal language is applied. Arguably, an approach based on the randomization-centered criterion without directly confronting the difficulties listed earlier would be possible only if randomized trials with no major flaws were the only experimental studies under consideration, in which case cautions about causal interpretation

could be reserved only for observational studies. Randomization strengthens the plausibility of a causal interpretation of study results, but randomization alone is not sufficient. Conversely, the absence of randomization does not on its own render a causal interpretation completely untenable. For observational studies, the blanket prohibition of causal language skirts the difficult but necessary work of judging whether a causal interpretation of any specific observational analysis is tenable. This judgment cannot rest on simply noting the absence of randomization¹⁵⁵; it requires context-informed examination of all relevant aspects of design, conduct, and analysis.

An Alternative Framework for Causal Inference for Medical and Health Policy Research

The extensive literature on causal inference across diverse disciplines^{26,30,156-158} suggests an alternative framework for observational studies that aim to answer questions about the causal effects of interventions. This framework avoids the limitations discussed earlier and can help editors and readers determine whether a particular observational study provides valid and reliable evidence about the effects of interventions in a target population. Such a framework can be summarized in terms of several core questions that need to be considered to understand and interpret observational studies:

1. **What is the causal question?** If the goal of the research is to provide evidence about the effects of medical or health policy interventions, the research question is best explicitly framed in causal terms, comparing 2 or more well-defined alternatives with respect to clearly defined outcomes of interest, for a specific target population during a period of follow-up.^{159,160}
2. **What quantity would, if known, answer the causal question?** After stating the causal question, one can specify the quantity that could, if known, serve as the answer to the question; this quantity is the *causal estimand* (eg, the causal effect of interest).^{161,162} The precise specification of the causal estimand requires describing the population of interest, the interventions or strategies to be compared, details of outcome definitions and the timing of outcome ascertainment, and the choice of effect measure (eg, risk difference, relative risk). The causal estimand can be formally specified using mathematical causal models (eg, closely related counterfactual, potential outcome, or structural models^{3,5,26,163-169}). In many cases, specification can be aided by describing the (hypothetical) target trial that could address the research question.^{144,170-172}
3. **What is the study design?** The approach for collecting new data or using existing data—including choosing among data sources, sampling individuals and their follow-up experience, and collecting treatment covariate and outcome information over time—determines whether the data can be used to answer the causal question. For example, in cohort studies comparing different treatment strategies, the choice of the start of follow-up (time zero) and the alignment of that time with the time at which eligibility is determined can affect the validity of observational analyses.¹⁷³ More broadly, the key goal of study design is to make the causal assumptions more plausible and to facilitate learning about the causal estimand.
4. **What causal assumptions are being made?** Drawing causal inferences from observational studies requires causal assumptions that allow investigators to learn about the causal estimand by using data. For example, many observational studies require an assumption that, given the variables that have been measured and accounted for (via study design or analysis), there remains no uncontrolled confounding.¹⁷⁴ Other approaches, such as instrumental variable analyses, difference-in-differences analyses, or regression discontinuity analyses, require different sets of assumptions. Typically, causal assumptions are untestable in the sense that they cannot be fully evaluated with the data alone; instead, they have to be examined on the basis of background knowledge (eg, clinical knowledge of the treatment selection process).^{175,176}
5. **How can the observed data be used to answer the causal question in principle and in practice?** Using the study design and causal assumptions, investigators can determine how analyses of observed data could, at least in principle (eg, if, hypothetically, all causal assumptions held and sampling variability were absent), provide information about the causal estimand. The formal examination of whether the observed data can in principle be used to learn about the causal estimand is referred to as *identification analysis*. In some cases, the assumptions suffice only to place bounds around the causal estimand.^{45,177-179} Most studies aiming to estimate causal estimands using observational data rely on well-understood *identification strategies* (ie, the results from prior identification analyses)^{180,181} and apply statistical methods to data for estimation and statistical inference. We offer a more detailed description of the relationship between causal estimands, identification analysis, and the use of data and statistical methods in the eText; eFigure 1, eFigure 2, and eFigure 3; and Example 1 and Example 2 in the [Supplement](#).
The statistical methods for observational studies should have good statistical performance (eg, acceptably low bias, high precision) and support the valid quantification of uncertainty (eg, producing valid CIs). The challenges of drawing statistical inferences using data and models are, if anything, accentuated in nonexperimental research.¹⁸² Furthermore, issues related to missing data and measurement error often arise in observational studies and require additional assumptions (typically untestable using the data alone) about the structure of missingness or measurement error, additional data (eg, validation studies), and specialized methods to address these issues and properly quantify uncertainty.
6. **Is a causal interpretation of the analyses tenable?** Evaluating the appropriateness of endowing the results of an observational analysis with a causal interpretation typically requires untestable assumptions. Determining whether such interpretation is tenable, therefore, involves subjective judgments informed by background knowledge and an understanding of the research context, drawing on multiple sources of evidence. These judgments can be informed by triangulation of results across different analyses (eg, using different assumptions or other data sources)¹⁸³; attempts to falsify the causal assumptions with the data, when possible (eg, negative control analyses^{80,184}); and quantitative bias/sensitivity analyses and other methods to examine assumption violations.^{19,185-191}

What This Framework Aims to Accomplish

This framework maintains the distinction between causation and association while addressing the limitations of approaches that rely on randomization as the sole criterion: it differentiates between causal ends and the statistical means to achieve them; supports the alignment between causal questions and the analyses used to answer them; increases transparency to facilitate scientific conversations; acknowledges that subjective judgments, informed by background clinical or policy knowledge, are unavoidable in observational studies; and aims to instill intellectual humility. Disagreements regarding the appropriate interpretation of observational studies among different stakeholders are always possible. This framework clarifies such disagreements by making the relevant considerations explicit and facilitates reasoning and debate.

Far from being a list of separate items, the framework highlights that multiple interrelated components are needed to report, evaluate, and interpret observational studies. For example, investigators will select study designs that are tailored to answer the causal question of interest and that support the plausibility of the causal assumptions needed to answer it. Similarly, study design and data analysis aspects can be arranged to facilitate the conduct of quantitative bias/sensitivity and falsification analyses, providing for the rigorous evaluation of assumptions. Background knowledge and understanding of the medical or policy context of the investigation is needed in all steps of the framework, from framing the research question to evaluating the plausibility of assumptions and evaluating whether a causal interpretation is tenable.

Interpreted practically, the framework allows the use of causal language to specify research questions and study goals (eg, in a manuscript's Introduction section); to describe study methods, assumptions under which the methods produce results that have a causal interpretation, and approaches for examining assumptions (eg, in the Methods section); and to reason about the plausibility of assumptions and the degree to which a causal interpretation is tenable in view of background knowledge while acknowledging the potential limitations of such an interpretation (eg, in the Discussion section). Two elements are central to this proposal for presenting observational studies: first, being explicit about the "if-then" (conditional) structure needed for their interpretation (eg, if certain assumptions hold, then a causal interpretation of the findings is tenable); and second, acknowledging that careful context-informed judgments are necessary to evaluate whether assumptions are plausible and a causal interpretation is tenable.

Last, although not the focus of this communication, the framework can also be applied to randomized trials and may be particularly helpful for pragmatic trials with baseline randomization that otherwise share many characteristics of observational studies (eg, trials with nonstandardized follow-up protocols and limited systematic efforts to enhance adherence to the assigned treatment).^{192,193}

What the Framework Does Not Do

The framework does not imply that all, or even most, observational studies merit a causal interpretation. For some observational studies that start with causal goals, causal inference may prove impossible;

in these cases, estimates retain only associational interpretations. In addition, many important descriptive and predictive research questions can be answered by observational studies that do not require causal notions.

Furthermore, when addressing causal questions, our proposal does not single out any of the currently popular frameworks, empirical research strategies, or statistical methods for causal inference from observational studies (eg, structural approaches^{27,167}; identification strategies^{180,181}; the target trial framework^{144,170}; the causal roadmap and targeted learning^{32,156}; any specific statistical, epidemiologic, or econometric method), nor does it single out any philosophy of statistical inference (eg, frequentist, bayesian). There is room for creativity in approaching practical causal questions, and investigators should have the freedom to select the approaches that best suit their research questions, provided they follow the norms for reporting described earlier. Without delving into the details of a specific research question, perhaps the most that can be recommended is to use the simplest methods that are adequate for the study's causal goals.^{194,195}

The framework does not address the broader issue of how to determine whether some general causal claim is warranted (eg, whether some exposure is a "cause" of some outcome). Instead, it focuses on whether observational studies can contribute independent credible evidence about causal effects of interventions in a particular target population, time, and place. Reports of such studies are the core publication type in most medical and health policy journals; more important, they are a key input to the process of evidence synthesis that can support general causal claims. This process combines information from multiple sources, including—in addition to trials and observational studies comparing interventions—basic science investigations, case reports, noncomparative studies, meta-analyses, and simulation modeling studies, as well as background knowledge.

Last, the framework does not cover other important issues that apply broadly to empirical investigations regardless of study design, such as prespecifying and preregistering analyses, following the principles of reproducible science, and sharing research materials.

Implications for Authors, Reviewers, Editors, and Readers

Adoption and further elaboration of the framework outlined earlier by medical journals offer the promise of facilitating communication between authors, reviewers, editors, and readers, but come with challenges in operationalization and implementation.

For authors, the framework provides more freedom to express causal goals and assumptions of observational studies, but also entails the responsibility to explicitly discuss and evaluate assumptions and openly acknowledge limitations (eg, violations of assumptions) and may require additional work (eg, to report technical details; to conduct triangulation, falsification, and bias analyses).

For reviewers, the framework should aid in the assessment of manuscripts that report observational studies. It requires familiarity with causal inference methods, as well as background knowledge to judge the appropriateness of the methods in the context of applied work.

Adoption of the framework should facilitate communication between authors, reviewers, and editors by encouraging the

transparent reporting and critique of methods and results of observational studies of medical interventions. Implementation at scale will require retaining expert reviewers and increasing the cooperation between editors, authors, and reviewers to operationalize the framework for use with different analyses and specific clinical applications and to evaluate whether it improves the reporting of empirical research. Furthermore, the complex judgments that the framework entails require vigilance to mitigate cognitive biases and distortions that may influence the presentation and interpretation of observational studies, particularly those using technically complex methods.¹⁹⁶

For readers, the framework should facilitate the clear communication of causal questions and methods. As usual, detailed tech-

nical descriptions may be appropriately placed in supplemental appendices to allow for the inclusion of the necessary detail and to maintain the readability and accessibility of the published study. Although our proposal suggests that complex concepts and more elaborate methodological descriptions may be needed to fully report and evaluate observational studies, adoption of the framework promises to improve the value of applied research that can support medical and policy decisions.

We look forward to readers' reactions to the framework. In future communications, we plan to explore its application in the context of concrete examples of specific types of observational analyses typically encountered in medical journals such as *JAMA* and the *JAMA Network* journals.

ARTICLE INFORMATION

Accepted for Publication: April 15, 2024.

Published Online: May 9, 2024.

doi:10.1001/jama.2024.7741

Conflict of Interest Disclosures: Dr Dahabreh reported receiving grants from Sanofi as principal investigator of a research agreement between Harvard and Sanofi for causal inference methods for transportability analyses; and reported receiving consulting fees from Moderna for trial and observational analyses outside the submitted work. No other disclosures were reported.

Additional Contributions: We thank Caroline Sietmann, MALIS, *JAMA* and *JAMA Network*, for assistance with soliciting and organizing comments on earlier versions of this article. We thank the following individuals for comments on earlier versions: Heather Gwynn Allore, MS, PhD, Yale University and *JAMA Internal Medicine*; Joshua D. Angrist, PhD, Massachusetts Institute of Technology; Michael Berkwitz, MD; Jesse Berlin, ScD, Rutgers University and *JAMA Network Open*; Isabelle Boutron, MD, PhD, Université de Paris, CRESS, Inserm; Stephen R. Cole, PhD, University of North Carolina at Chapel Hill; John Concato, MD, MS, MPH, US Food and Drug Administration and Yale School of Medicine; Gregory Curfman, MD, *JAMA* and *JAMA Network*; Annette Flanagan, RN, MA, *JAMA* and *JAMA Network*; Maria Glymour, ScD, MS, Boston University; Deborah Grady, MD, MPH, University of California, San Francisco, and *JAMA Internal Medicine*; Sander Greenland, DrPH, University of California, Los Angeles; Gordon Guyatt, MD, MSc, McMaster University; Sebastian Haneuse, PhD, Harvard University and *JAMA Network Open*; Frank E. Harrell Jr, PhD, Vanderbilt University; Robert A. Harrington, MD, Weill Cornell Medicine and *JAMA Cardiology*; Laura Hatfield, PhD, Harvard University and *JAMA*; Miguel A. Hernán, MD, PhD, Harvard University; Guido Imbens, PhD, Stanford University; Nina Joyce, PhD, Brown University; Amy H. Kaji, MD, PhD, University of California, Los Angeles, and *JAMA Surgery*; Jay S. Kaufman, PhD, McGill University; Dhruv Kazi, MD, MSc, MS, Beth Israel Deaconess Medical Center; Kenneth S. Kendler, MD, Virginia Commonwealth University; Daniel B. Kramer, MD, MPH, Beth Israel Deaconess Medical Center; Timothy Lash, DSc, MPH, Emory University; Roger J. Lewis, MD, PhD, University of California, Los Angeles, and *JAMA*; Charles F. Manski, PhD, Northwestern University; M. Hassan Murad, MD, Mayo Clinic; Christopher Muth, MD, *JAMA*; Sharon-Lise Normand, PhD, Harvard University; Neil Pearce, PhD, London

School of Hygiene and Tropical Medicine; Maya Petersen, MD, PhD, University of California, Berkeley; Romain Pirracchio, MD, MPH, PhD, University of California, San Francisco, and *JAMA*; Stuart J. Pocock, PhD, London School of Hygiene and Tropical Medicine; James M. Robins, MD, Harvard University; Sherri Rose, PhD, Stanford University; Paul R. Rosenbaum, PhD, University of Pennsylvania; Kenneth J. Rothman, DrPH, Boston University; Jeffrey Saver, MD, University of California, Los Angeles, and *JAMA*; Stephen Schenkel, MD, MPP, University of Maryland Medical Center and *JAMA*; David Schriger, MD, MPH, University of California, Los Angeles, and *JAMA*; Ian Shrier, MD, PhD, McGill University; Dylan S. Small, PhD, University of Pennsylvania; George Davey Smith, DSc, University of Bristol; Zirui Song, MD, PhD, Harvard University and *JAMA Health Forum*; Jonathan A. C. Sterne, PhD, University of Bristol; Elizabeth A. Stuart, PhD, Johns Hopkins University and *JAMA Health Forum*; Sonja A. Swanson, ScD, University of Pittsburgh and *JAMA Psychiatry*; Eric Tchetgen Tchetgen, PhD, University of Pennsylvania; Linda Valeri, PhD, Columbia University and *JAMA Psychiatry*; Tyler J. VanderWeele, PhD, Harvard University and *JAMA Psychiatry*; Rishi Wadhwa, MD, MPP, MPhil, Beth Israel Deaconess Medical Center; Robert Yeh, MD, MS, MBA, Beth Israel Deaconess Medical Center; and Alan M. Zaslavsky, PhD, Harvard Medical School.

REFERENCES

1. Cochran WG. The planning of observational studies of human populations. *J R Stat Soc [Ser A]*. 1965;128(2):234-266. doi:10.2307/2344179
2. Susser M. *Causal Thinking in the Health Sciences: Concepts and Strategies of Epidemiology*. Oxford University Press; 1973:xii.
3. Rubin DB. Estimating causal effects of treatments in randomized and nonrandomized studies. *J Educ Psychol*. 1974;66(5):688-701. doi:10.1037/h0037350
4. Rubin DB. Bayesian inference for causal effects: the role of randomization. *Ann Stat*. 1978;6:34-58. doi:10.1214/aos/1176344064
5. Rubin DB. Randomization analysis of experimental data: the Fisher randomization test comment. *J Am Stat Assoc*. 1980;75(371):591-593. doi:10.2307/2287653
6. Rothman KJ, Greenland S, Walker AM. Concepts of interaction. *Am J Epidemiol*. 1980;112(4):467-470. doi:10.1093/oxfordjournals.aje.a113015

7. Cochran WG. *Planning and Analysis of Observational Studies*. John Wiley & Sons; 1983. doi:10.1002/9780470316542
8. Miettinen OS. *Theoretical Epidemiology: Principles of Occurrence Research in Medicine*. Delmar Publishers; 1985:xxii.
9. Robins J. A new approach to causal inference in mortality studies with a sustained exposure period—application to control of the healthy worker survivor effect. *Math Model*. 1986;7(9-12):1393-1512. doi:10.1016/0270-0255(86)90088-6
10. Greenland S, Robins JM. Identifiability, exchangeability, and epidemiological confounding. *Int J Epidemiol*. 1986;15(3):413-419. doi:10.1093/ije/15.3.413
11. Holland PW. Statistics and causal inference. *J Am Stat Assoc*. 1986;81(396):945-960. doi:10.1080/01621459.1986.10478354
12. Robins JM, Morgenstern H. The foundations of confounding in epidemiology. *Comput Math Appl*. 1987;14(9-12):869-916. doi:10.1016/0898-1221(87)90236-7
13. Greenland S. Randomization, statistics, and causal inference. *Epidemiology*. 1990;1(6):421-429. doi:10.1097/00001648-199011000-00003
14. Robins JM, Greenland S. Identifiability and exchangeability for direct and indirect effects. *Epidemiology*. 1992;3(2):143-155. doi:10.1097/00001648-199203000-00013
15. Cox DR. Causality: some statistical aspects. *J R Stat Soc Ser A Stat Soc*. 1992;155(2):291-301. doi:10.2307/2982962
16. Halloran ME, Struchiner CJ. Causal inference in infectious diseases. *Epidemiology*. 1995;6(2):142-151. doi:10.1097/00001648-199503000-00010
17. Little RJ, Rubin DB. Causal effects in clinical and epidemiological studies via potential outcomes: concepts and analytical approaches. *Annu Rev Public Health*. 2000;21(1):121-145. doi:10.1146/annurev.publhealth.21.1.121
18. Shadish WR, Cook TD, Campbell DT. *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton Mifflin; 2001:xxi.
19. Rosenbaum P. *Observational Studies*. Springer-Verlag; 2002. doi:10.1007/978-1-4757-3692-2
20. Frangakis CE, Rubin DB. Principal stratification in causal inference. *Biometrics*. 2002;58(1):21-29. doi:10.1111/j.0006-341X.2002.00021.x

21. Hernán MA. A definition of causal effect for epidemiological research. *J Epidemiol Community Health*. 2004;58(4):265-271. doi:10.1136/jech.2002.006361
22. Rothman KJ, Greenland S. Causation and causal inference in epidemiology. *Am J Public Health*. 2005;95(suppl 1):S144-S150. doi:10.2105/AJPH.2004.059204
23. Hernán MA, Robins JM. Estimating causal effects from epidemiological data. *J Epidemiol Community Health*. 2006;60(7):578-586. doi:10.1136/jech.2004.029496
24. Heckman JJ. Econometric causality. *Int Stat Rev*. 2008;76(1):1-27. doi:10.1111/j.1751-5823.2007.00024.x
25. Hudgens MG, Halloran ME. Toward causal inference with interference. *J Am Stat Assoc*. 2008;103(482):832-842. doi:10.1198/016214508000000292
26. Pearl J. Causal inference in statistics: an overview. *Statist Serv*. 2009;3:96-146.
27. Pearl J. *Causality*. Cambridge University Press; 2009. doi:10.1017/CBO9780511803161
28. Angrist JD, Pischke JS. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press; 2009. doi:10.1515/9781400829828
29. VanderWeele TJ. On the distinction between interaction and effect modification. *Epidemiology*. 2009;20(6):863-871. doi:10.1097/EDE.0b013e3181ba333c
30. Angrist JD, Pischke JS. The credibility revolution in empirical economics: how better research design is taking the con out of econometrics. *J Econ Perspect*. 2010;24(2):3-30. doi:10.1257/jep.24.2.3
31. Rosenbaum PR. *Design of Observational Studies*. Springer; 2010. doi:10.1007/978-1-4419-1213-8
32. Van der Laan MJ, Rose S. *Targeted Learning: Causal Inference for Observational and Experimental Data*. Springer; 2011. doi:10.1007/978-1-4419-9782-1
33. Berzuini C, Dawid P, Bernardinelli L, eds. *Causality: Statistical Perspectives and Applications*. John Wiley & Sons; 2012. doi:10.1002/9781119945710
34. Imbens GW, Rubin DB. *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge University Press; 2015. doi:10.1017/CBO9781139025751
35. Athey S, Imbens GW. The state of applied econometrics: causality and policy evaluation. *J Econ Perspect*. 2017;31(2):3-32. doi:10.1257/jep.31.2.3
36. Young JG, Stensrud MJ, Tchetgen Tchetgen EJ, Hernán MA. A causal framework for classical statistical estimands in failure-time settings with competing events. *Stat Med*. 2020;39(8):1199-1236. doi:10.1002/sim.8471
37. Hernán MA, Robins JM. Causal inference: what if. Accessed April 24, 2024. <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>
38. Imbens GW. Causal inference in the social sciences. Accessed April 24, 2024. doi:10.1146/annurev-statistics-033121-114601
39. Greenland S, Pearl J, Robins JM. Confounding and collapsibility in causal inference. *Stat Sci*. 1999;14(1):29-46. doi:10.1214/ss/1009211805
40. Greenland S, Brumback B. An overview of relations among causal modelling methods. *Int J Epidemiol*. 2002;31(5):1030-1037. doi:10.1093/ije/31.5.1030
41. Laan MJ, Robins JM. *Unified Methods for Censored Longitudinal Data and Causality*. Springer; 2003. doi:10.1007/978-0-387-21700-0
42. Cochran WG, Rubin DB. Controlling bias in observational studies: a review. *Sankhya Ser A*. 1973;35(4):417-446.
43. Rosenbaum PR, Rubin DB. The central role of the propensity score in observational studies for causal effects. *Biometrika*. 1983;70(1):41-55. doi:10.1093/biomet/70.1.41
44. Rosenbaum PR, Rubin DB. Reducing bias in observational studies using subclassification on the propensity score. *J Am Stat Assoc*. 1984;79(387):516-524. doi:10.1080/01621459.1984.10478078
45. Manski CF. Nonparametric bounds on treatment effects. *Am Econ Rev*. 1990;80(2):319-323.
46. Angrist J, Imbens G. *Identification and Estimation of Local Average Treatment Effects*. National Bureau of Economic Research Cambridge; 1995. doi:10.3386/t0118
47. Angrist JD, Imbens GW, Rubin DB. Identification of causal effects using instrumental variables. *J Am Stat Assoc*. 1996;91(434):444-455. doi:10.1080/01621459.1996.10476902
48. Hahn J. On the role of the propensity score in efficient semiparametric estimation of average treatment effects. *Econometrica*. 1998;66(2):315-331. doi:10.2307/2998560
49. Robins JM. Association, causation, and marginal structural models. *Synthese*. 1999;121(1-2):151-179. doi:10.1023/A:1005285815569
50. Hahn J, Todd P, Van der Klaauw W. Identification and estimation of treatment effects with a regression-discontinuity design. *Econometrica*. 2001;69(1):201-209. doi:10.1111/1468-0262.00183
51. Hirano K, Imbens GW, Ridder G. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*. 2003;71(4):1161-1189. doi:10.1111/1468-0262.00442
52. Imbens GW. Nonparametric estimation of average treatment effects under exogeneity: a review. *Rev Econ Stat*. 2004;86(1):4-29. doi:10.1162/003465304323023651
53. Lunceford JK, Davidian M. Stratification and weighting via the propensity score in estimation of causal treatment effects: a comparative study. *Stat Med*. 2004;23(19):2937-2960. doi:10.1002/sim.1903
54. Bang H, Robins JM. Doubly robust estimation in missing data and causal inference models. *Biometrics*. 2005;61(4):962-973. doi:10.1111/j.1541-0420.2005.00377.x
55. Rubin DB. *Matched Sampling for Causal Effects*. Cambridge University Press; 2006. doi:10.1017/CBO9780511810725
56. Hernán MA, Robins JM. Instruments for causal inference: an epidemiologist's dream? *Epidemiology*. 2006;17(4):360-372. doi:10.1097/01.ede.0000222409.00878.37
57. Robins JM, Hernán MA, Brumback B. Marginal structural models and causal inference in epidemiology. *Epidemiology*. 2000;11(5):550-560. doi:10.1097/00001648-200009000-00011
58. Murphy SA. Optimal dynamic treatment regimes. *J R Stat Soc Series B Stat Methodol*. 2003;65(2):331-355. doi:10.1111/1467-9868.00389
59. Gelman A, Meng XL, eds. *Applied Bayesian Modeling and Causal Inference From Incomplete-Data Perspectives*. John Wiley & Sons; 2004. doi:10.1002/0470090456
60. Athey S, Imbens GW. Identification and inference in nonlinear difference-in-differences models. *Econometrica*. 2006;74(2):431-497. doi:10.1111/j.1468-0262.2006.00668.x
61. Ho DE, Imai K, King G, Stuart EA. Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference. *Polit Anal*. 2007;15(3):199-236. doi:10.1093/pan/mpi013
62. Imai K, King G, Stuart EA. Misunderstandings between experimentalists and observationalists about causal inference. *J R Stat Soc Ser A Stat Soc*. 2008;171(2):481-502. doi:10.1111/j.1467-985X.2007.00527.x
63. Robins JM, Hernán MA. *Estimation of the Causal Effects of Time-Varying Exposures: Longitudinal Data Analysis*. Chapman & Hall/CRC; 2008:547-593. doi:10.1201/9781420011579.ch23
64. Imbens GW, Lemieux T. Regression discontinuity designs: a guide to practice. *J Econom*. 2008;142(2):615-635. doi:10.1016/j.jeconom.2007.05.001
65. Sekhon JS. Opiates for the matches: matching methods for causal inference. *Annu Rev Polit Sci*. 2009;12:487-508. doi:10.1146/annurev.polisci.11.060606.135444
66. Stuart EA. Matching methods for causal inference: a review and a look forward. *Stat Sci*. 2010;25(1):1-21. doi:10.1214/09-STS313
67. Lee DS, Lemieux T. Regression discontinuity designs in economics. *J Econ Lit*. 2010;48(2):281-355. doi:10.1257/jel.48.2.281
68. Orellana L, Rotnitzky A, Robins JM. Dynamic regime marginal structural mean models for estimation of optimal dynamic treatment regimes, part I: main content. *Int J Biostat*. 2010;6(2):8. doi:10.2202/1557-4679.1200
69. Orellana L, Rotnitzky A, Robins JM. Dynamic regime marginal structural mean models for estimation of optimal dynamic treatment regimes, part II: proofs of results. *Int J Biostat*. 2010;6(2):9. doi:10.2202/1557-4679.1242
70. Abadie A, Diamond A, Hainmueller J. Synthetic control methods for comparative case studies: estimating the effect of California's tobacco control program. *J Am Stat Assoc*. 2010;105(490):493-505. doi:10.1198/jasa.2009.ap08746
71. Lechner M. The estimation of causal effects by difference-in-difference methods. *Found Trends Econom*. 2011;4(3):165-224. doi:10.1561/08000000014
72. Hill JL. Bayesian nonparametric modeling for causal inference. *J Comput Graph Stat*. 2011;20(1):217-240. doi:10.1198/jcgs.2010.08162
73. Tchetgen Tchetgen EJ, VanderWeele TJ. On causal inference in the presence of interference. *Stat Methods Med Res*. 2012;21(1):55-75. doi:10.1177/0962280210386779
74. Valeri L, Vanderweele TJ. Mediation analysis allowing for exposure-mediator interactions and causal interpretation: theoretical assumptions and

- implementation with SAS and SPSS macros. *Psychol Methods*. 2013;18(2):137-150. doi:10.1037/a0031034
75. Chakraborty B, Moodie EEM. *Statistical Methods for Dynamic Treatment Regimes: Reinforcement Learning, Causal Inference, and Personalized Medicine*. Springer; 2013. doi:10.1007/978-1-4614-7428-9
76. VanderWeele TJ, Knol MJ. A tutorial on interaction. *Epidemiol Methods*. 2014;3(1):33-72. doi:10.1515/em-2013-0005
77. VanderWeele T. *Explanation in Causal Inference: Methods for Mediation and Interaction*. Oxford University Press; 2015.
78. Athey S, Imbens GW. Machine learning methods for estimating heterogeneous causal effects. Published April 2015. Accessed April 23, 2024. <https://gsb-faculty.stanford.edu/guido-w-imbens/files/2022/04/3350.pdf>
79. Abadie A, Imbens GW. Matching on the estimated propensity score. *Econometrica*. 2016;84(2):781-807. doi:10.3982/ECTA11293
80. Sofer T, Richardson DB, Colicino E, Schwartz J, Tchetgen Tchetgen EJ. On negative outcome control of unobserved confounding as a generalization of difference-in-differences. *Stat Sci*. 2016;31(3):348-361. doi:10.1214/16-ST558
81. Cain LE, Robins JM, Lanoy E, Logan R, Costagliola D, Hernán MA. When to start treatment? a systematic approach to the comparison of dynamic regimes using observational data. *Int J Biostat*. 2010;6(2):18. doi:10.2202/1557-4679.1212
82. Abadie A, Cattaneo MD. Econometric methods for program evaluation. *Annu Rev Econ*. 2018;10:465-503. doi:10.1146/annurev-economics-080217-053402
83. Chernozhukov V, Chetverikov D, Demirer M, et al. Double/debiased machine learning for treatment and structural parameters. *Econom J*. 2018;21(1):C1-C68. doi:10.1111/ectj.12097
84. Miao W, Geng Z, Tchetgen Tchetgen E. Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*. 2018;105(4):987-993. doi:10.1093/biomet/asv038
85. Wager S, Athey S. Estimation and inference of heterogeneous treatment effects using random forests. *J Am Stat Assoc*. 2018;113(523):1228-1242. doi:10.1080/01621459.2017.1319839
86. Li F, Morgan KL, Zaslavsky AM. Balancing covariates via propensity score weighting. *J Am Stat Assoc*. 2018;113(521):390-400. doi:10.1080/01621459.2016.1260466
87. Künzel SR, Sekhon JS, Bickel PJ, Yu B. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proc Natl Acad Sci U S A*. 2019;116(10):4156-4165. doi:10.1073/pnas.1804597116
88. Cattaneo MD, Idrobo N, Titiunik R. *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press; 2019. doi:10.1017/9781108684606
89. Tsiatis AA. *Dynamic Treatment Regimes: Statistical Methods for Precision Medicine*. CRC Press; 2019. doi:10.1201/9780429192692
90. Fröhlich M, Frölich M, Sperlich S. *Impact Evaluation*. Cambridge University Press; 2019. doi:10.1017/9781107337008
91. Abadie A. Using synthetic controls: feasibility, data requirements, and methodological aspects. *J Econ Lit*. 2021;59(2):391-425. doi:10.1257/jel.20191450
92. Brumback BA. *Fundamentals of Causal Inference: With R*. Chapman & Hall/CRC; 2021. doi:10.1201/9781003146674
93. Forastiere L, Airolidi EM, Mealli F. Identification and estimation of treatment and interference effects in observational studies on networks. *J Am Stat Assoc*. 2021;116(534):901-918. doi:10.1080/01621459.2020.1768100
94. Cattaneo MD, Titiunik R. Regression discontinuity designs. *Annu Rev Econ*. 2022;14:821-851. doi:10.1146/annurev-economics-051520-021409
95. Roth J, Sant'Anna PH, Bilinski A, Poe J. What's trending in difference-in-differences? a synthesis of the recent econometrics literature. *J Econom*. 2023;235(2):2218-2244. doi:10.1016/j.jeconom.2023.03.008
96. Huber M. *Causal Analysis: Impact Evaluation and Causal Machine Learning With Applications in R*. MIT Press; 2023.
97. Cattaneo MD, Idrobo N, Titiunik R. *A Practical Introduction to Regression Discontinuity Designs: Extensions*. Cambridge University Press; 2024. doi:10.1017/9781009441896
98. Ogburn EL, Sofrygin O, Diaz I, Van der Laan MJ. Causal inference for social network data. *J Am Stat Assoc*. 2024;119(545):597-611. doi:10.1080/01621459.2022.2131557
99. Rosenbaum PR. Model-based direct adjustment. *J Am Stat Assoc*. 1987;82(398):387-394. doi:10.1080/01621459.1987.10478441
100. Drake C. Effects of misspecification of the propensity score on estimators of treatment effect. *Biometrics*. 1993;49:1231-1236. doi:10.2307/2532266
101. Imbens G, Angrist J. Identification and estimation of local average treatment effects. *Econometrica*. 1994;61(2):467-476. doi:10.2307/2951620
102. Heckman JJ, Ichimura H, Todd PE. Matching as an econometric evaluation estimator: evidence from evaluating a job training programme. *Rev Econ Stud*. 1997;64(4):605-654. doi:10.2307/2971733
103. Heckman JJ, Vytlacil EJ. Local instrumental variables and latent variable models for identifying and bounding treatment effects. *Proc Natl Acad Sci U S A*. 1999;96(8):4730-4734. doi:10.1073/pnas.96.8.4730
104. Greenland S. An introduction to instrumental variables for epidemiologists. *Int J Epidemiol*. 2000;29(4):722-729. doi:10.1093/ije/29.4.722
105. Rosenbaum PR. Covariance adjustment in randomized experiments and observational studies. *Stat Sci*. 2002;17(3):286-327. doi:10.1214/ss/1042727942
106. Maldonado G, Greenland S. Estimating causal effects. *Int J Epidemiol*. 2002;31(2):422-429. doi:10.1093/ije/31.2.422
107. Bertrand M, Duflo E, Mullainathan S. How much should we trust differences-in-differences estimates? *Q J Econ*. 2004;119(1):249-275. doi:10.1162/003355304772839588
108. Lee MJ. *Micro-Econometrics for Policy, Program and Treatment Effects*. OUP Oxford; 2005. doi:10.1093/0199267693.001.0001
109. Abadie A. Semiparametric difference-in-differences estimators. *Rev Econ Stud*. 2005;72(1):1-19. doi:10.1111/0034-6527.00321
110. Abadie A, Imbens GW. Large sample properties of matching estimators for average treatment effects. *Econometrica*. 2006;74(1):235-267. doi:10.1111/j.1468-0262.2006.00655.x
111. Van Der Laan MJ, Rubin D. Targeted maximum likelihood learning. *Int J Biostat*. 2006;2:1. doi:10.2202/1557-4679.1043
112. Petersen ML, Sinisi SE, van der Laan MJ. Estimation of direct causal effects. *Epidemiology*. 2006;17(3):276-284. doi:10.1097/01.ede.0000208475.99429.2d
113. Donald SG, Lang K. Inference with difference-in-differences and other panel data. *Rev Econ Stat*. 2007;89(2):221-233. doi:10.1162/rest.89.2.221
114. Imbens GW, Wooldridge JM. Recent developments in the econometrics of program evaluation. *J Econ Lit*. 2009;47(1):5-86. doi:10.1257/jel.47.1.5
115. VanderWeele TJ. Marginal structural models for the estimation of direct and indirect effects. *Epidemiology*. 2009;20(1):18-26. doi:10.1097/EDE.0b013e31818f69ce
116. van der Laan MJ, Gruber S. Collaborative double robust targeted maximum likelihood estimation. *Int J Biostat*. 2010;6(1):17. doi:10.2202/1557-4679.1181
117. Austin PC. An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behav Res*. 2011;46(3):399-424. doi:10.1080/00273171.2011.568786
118. Hainmueller J. Entropy balancing for causal effects: a multivariate reweighting method to produce balanced samples in observational studies. *Polit Anal*. 2012;20(1):25-46. doi:10.1093/pan/mpm025
119. Tchetgen Tchetgen EJ, Shpitser I. Semiparametric theory for causal mediation analysis: efficiency bounds, multiple robustness, and sensitivity analysis. *Ann Stat*. 2012;40(3):1816-1845. doi:10.1214/12-AOS990
120. Daniel RM, Cousens SN, De Stavola BL, Kenward MG, Sterne JAC. Methods for dealing with time-dependent confounding. *Stat Med*. 2013;32(9):1584-1618. doi:10.1002/sim.5686
121. Austin PC. The performance of different propensity score methods for estimating marginal hazard ratios. *Stat Med*. 2013;32(16):2837-2849. doi:10.1002/sim.5705
122. Belloni A, Chernozhukov V, Hansen C. High-dimensional methods and inference on structural and treatment effects. *J Econ Perspect*. 2014;28(2):29-50. doi:10.1257/jep.28.2.29
123. Baiocchi M, Cheng J, Small DS. Instrumental variable methods for causal inference. *Stat Med*. 2014;33(13):2297-2340. doi:10.1002/sim.6128
124. Farrell MH. Robust inference on average treatment effects with possibly more covariates than observations. *J Econom*. 2015;189(1):1-23. doi:10.1016/j.jeconom.2015.06.017
125. Zubizarreta JR. Stable weights that balance covariates for estimation with incomplete outcome data. *J Am Stat Assoc*. 2015;110(511):910-922. doi:10.1080/01621459.2015.1023805

126. VanderWeele TJ. Mediation analysis: a practitioner's guide. *Annu Rev Public Health*. 2016; 37:17-32. doi:10.1146/annurev-publhealth-032315-021402
127. Mattei A, Mealli F. Regression discontinuity designs as local randomized experiments. *Obs Stud*. 2016;2:156-173.
128. Ding P, VanderWeele TJ. Sensitivity analysis without assumptions. *Epidemiology*. 2016;27(3): 368-377. doi:10.1097/EDE.0000000000000457
129. VanderWeele TJ, Tchetgen Tchetgen EJ. Mediation analysis with time varying exposures and mediators. *J R Stat Soc Series B Stat Methodol*. 2017;79(3):917-938. doi:10.1111/rssb.12194
130. Belloni A, Chernozhukov V, Fernández-Val I, Hansen C. Program evaluation and causal inference with high-dimensional data. *Econometrica*. 2017; 85(1):233-298. doi:10.3982/ECTA12723
131. Ding P, Li F. Causal inference: a missing data perspective. *Stat Sci*. 2018;33(2):214-237. doi:10.1214/18-ST5645
132. Sant'Anna PHC, Zhao J. Doubly robust difference-in-differences estimators. *J Econom*. 2020;219(1):101-122. doi:10.1016/j.jeconom.2020.06.003
133. Hahn PR, Murray JS, Carvalho CM. Bayesian regression tree models for causal inference: regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Anal*. 2020;15 (3):965-1056. doi:10.1214/19-BA1195
134. Arkhangelsky D, Athey S, Hirshberg DA, Imbens GW, Wager S. Synthetic difference-in-differences. *Am Econ Rev*. 2021;111(12):4088-4118. doi:10.1257/aer.20190159
135. Callaway B, Sant'Anna P. Difference-in-differences with multiple time periods. *J Econom*. 2021;225(2):200-230. doi:10.1016/j.jeconom.2020.12.001
136. Ben-Michael E, Feller A, Rothstein J. The augmented synthetic control method. Accessed April 24, 2024. https://www.nber.org/system/files/working_papers/w28885/w28885.pdf doi:10.3386/w28885
137. Athey S, Bayate M, Doudchenko N, Imbens G, Khosravi K. Matrix completion methods for causal panel data models. *J Am Stat Assoc*. 2021;116:1-15. doi:10.1080/01621459.2021.1891924
138. Li F, Ding P, Mealli F. Bayesian causal inference: a critical review. *Philos Trans A Math Phys Eng Sci*. 2023;381(2247):20220153.
139. Christiansen S, Iverson C, Flanagan A, et al. *AMA Manual of Style: A Guide for Authors and Editors*. 11th ed. Oxford University Press; 2020.
140. Zaslavsky AM. Exploring potential causal inference through natural experiments. *JAMA Health Forum*. 2021;2(6):e210289. doi:10.1001/jamahealthforum.2021.0289
141. VanderWeele TJ. Can sophisticated study designs with regression analyses of observational data provide causal inferences? *JAMA Psychiatry*. 2021;78(3):244-246. doi:10.1001/jamapsychiatry.2020.2588
142. Haukoos JS, Lewis RJ. The propensity score. *JAMA*. 2015;314(15):1637-1638. doi:10.1001/jama.2015.13480
143. Lipsky AM, Greenland S. Causal directed acyclic graphs. *JAMA*. 2022;327(11):1083-1084. doi:10.1001/jama.2022.1816
144. Hernán MA, Wang W, Leaf DE. Target trial emulation: a framework for causal inference from observational data. *JAMA*. 2022;328(24):2446-2447. doi:10.1001/jama.2022.21383
145. Thomas L, Li F, Pencina M. Using propensity score methods to create target populations in observational clinical research. *JAMA*. 2020;323(5): 466-467. doi:10.1001/jama.2019.21558
146. Niknam BA, Zubizarreta JR. Using cardinality matching to design balanced and representative samples for observational studies. *JAMA*. 2022;327 (2):173-174. doi:10.1001/jama.2021.20555
147. Thomas LE, Li F, Pencina MJ. Overlap weighting: a propensity score method that mimics attributes of a randomized clinical trial. *JAMA*. 2020;323(23):2417-2418. doi:10.1001/jama.2020.7819
148. Maciejewski ML, Basu A. Regression discontinuity design. *JAMA*. 2020;324(4):381-382. doi:10.1001/jama.2020.3822
149. Maciejewski ML, Dowd BE, Norton EC. Instrumental variables and heterogeneous treatment effects. *JAMA*. 2022;327(12):1177-1178. doi:10.1001/jama.2022.2505
150. Emdin CA, Khera AV, Kathiresan S. Mendelian randomization. *JAMA*. 2017;318(19):1925-1926. doi:10.1001/jama.2017.17219
151. Dimick JB, Ryan AM. Methods for evaluating changes in health care policy: the difference-in-differences approach. *JAMA*. 2014;312 (22):2401-2402. doi:10.1001/jama.2014.16153
152. Rothman K. *Modern Epidemiology*. Little Brown & Co; 1986:77.
153. Savitz DARE. Re: "Associations are not effects". *Am J Epidemiol*. 1991;134(4):442-444. doi:10.1093/oxfordjournals.aje.a116110
154. Hernán MA. The C-word: scientific euphemisms do not improve causal inference from observational data. *Am J Public Health*. 2018;108 (5):616-619. doi:10.2105/AJPH.2018.304337
155. Rosenbaum PR. Discussing hidden bias in observational studies. *Ann Intern Med*. 1991;115(11): 901-905. doi:10.7326/0003-4819-115-11-901
156. Petersen ML, van der Laan MJ. Causal models and learning from data: integrating causal modeling and statistical estimation. *Epidemiology*. 2014;25 (3):418-426. doi:10.1097/EDE.0000000000000078
157. Glass TA, Goodman SN, Hernán MA, Samet JM. Causal inference in public health. *Annu Rev Public Health*. 2013;34:61-75. doi:10.1146/annurev-publhealth-031811-124606
158. Dang LE, Gruber S, Lee H, et al. A causal roadmap for generating high-quality real-world evidence. *J Clin Transl Sci*. 2023;7(1):e212. doi:10.1017/cts.2023.635
159. Guyatt G, Meade MQ, Achoris T, Richardson WS, Jaeschke R. What is the question? In: Guyatt G, Rennie D, Meade MQ, Cook DJ, eds. *Users' Guides to the Medical Literature: A Manual for Evidence-Based Clinical Practice*. 3rd ed. McGraw-Hill; 2008:17-28.
160. Stoto M, Oakes M, Stuart E, Savitz L, Priest EL, Zurovac J. Analytical methods for a learning health system. 1: framing the research question. *EGEMS (Wash DC)*. 2017;5(1):28. doi:10.5334/egems.250
161. US Department of Health and Human Services; Food and Drug Administration; Center for Drug Evaluation and Research (CDER); Center for Biologics Evaluation and Research (CBER). E9(R1) statistical principles for clinical trials: addendum: estimands and sensitivity analysis in clinical trials: guidance for industry. Accessed April 24, 2024. <https://www.fda.gov/media/148473/download>
162. Little RJ, Lewis RJ. Estimands, estimators, and estimates. *JAMA*. 2021;326(10):967-968. doi:10.1001/jama.2021.2886
163. Dawid AP. Causal inference without counterfactuals. *J Am Stat Assoc*. 2000;95(450): 407-424. doi:10.1080/01621459.2000.10474210
164. Robins JM, Greenland S. Causal inference without counterfactuals: comment. *J Am Stat Assoc*. 2000;95(450):431-435. doi:10.1080/01621459.2000.10474214
165. Pearl J. Causal diagrams for empirical research. *Biometrika*. 1995;82(4):669-688. doi:10.1093/biomet/82.4.669
166. Greenland S, Pearl J, Robins JM. Causal diagrams for epidemiologic research. *Epidemiology*. 1999;10(1):37-48. doi:10.1097/00001648-199901000-00008
167. Richardson TS, Robins JM. Single world intervention graphs (SWIGs): a unification of the counterfactual and graphical approaches to causality. Center for Statistics and the Social Sciences working paper 128. April 2013. Accessed April 23, 2024. <https://csss.uw.edu/research/working-papers/single-world-intervention-graphs-swigs-unification-counterfactual-and>
168. Spirtes P, Glymour CN, Scheines R. *Causation, Prediction, and Search*. MIT Press; 2000.
169. Ogburn EL, VanderWeele TJ. Causal diagrams for interference. *Stat Sci*. 2014;29(4):559-578. doi:10.1214/14-ST5501
170. Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *Am J Epidemiol*. 2016;183(8):758-764. doi:10.1093/aje/kwv254
171. Rubin DB. The design versus the analysis of observational studies for causal effects: parallels with the design of randomized trials. *Stat Med*. 2007;26(1):20-36. doi:10.1002/sim.2739
172. Rubin DB. For objective causal inference, design trumps analysis. *Ann Appl Stat*. 2008;2(3): 808-840. doi:10.1214/08-AOAS187
173. Hernán MA, Sauer BC, Hernández-Díaz S, Platt R, Shrier I. Specifying a target trial prevents immortal time bias and other self-inflicted injuries in observational analyses. *J Clin Epidemiol*. 2016;79: 70-75. doi:10.1016/j.jclinepi.2016.04.014
174. VanderWeele TJ. Principles of confounder selection. *Eur J Epidemiol*. 2019;34(3):211-219. doi:10.1007/s10654-019-00494-6
175. Robins JM. Data, design, and background knowledge in etiologic inference. *Epidemiology*. 2001;12(3):313-320. doi:10.1097/00001648-200105000-00011
176. Hernán MA, Hernández-Díaz S, Werler MM, Mitchell AA. Causal knowledge as a prerequisite for confounding evaluation: an application to birth defects epidemiology. *Am J Epidemiol*. 2002;155(2): 176-184. doi:10.1093/aje/k155.2.176
177. Manski CF. *Partial Identification of Probability Distributions*. Springer; 2003.
178. Balke A, Pearl J. Bounds on treatment effects from studies with imperfect compliance. *J Am Stat*

- Assoc. 1997;92(439):1171-1176. doi:[10.1080/01621459.1997.10474074](https://doi.org/10.1080/01621459.1997.10474074)
- 179.** Manski CF. *Patient Care Under Uncertainty*. Princeton University Press; 2019.
- 180.** Imbens GW. Potential outcome and directed acyclic graph approaches to causality: relevance for empirical practice in economics. *J Econ Lit*. 2020;58(4):1129-1179. doi:[10.1257/jel.20191597](https://doi.org/10.1257/jel.20191597)
- 181.** Angrist JD, Krueger AB. Empirical strategies in labor economics. In: Ashenfelter OC, Card D, eds. *Handbook of Labor Economics*. Vol 3. Elsevier; 1999: 1277-1366.
- 182.** Greenland S, Senn SJ, Rothman KJ, et al. Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations. *Eur J Epidemiol*. 2016;31(4):337-350. doi:[10.1007/s10654-016-0149-3](https://doi.org/10.1007/s10654-016-0149-3)
- 183.** Lawlor DA, Tilling K, Davey Smith G. Triangulation in aetiological epidemiology. *Int J Epidemiol*. 2016;45(6):1866-1886.
- 184.** Lipsitch M, Tchetgen Tchetgen E, Cohen T. Negative controls: a tool for detecting confounding and bias in observational studies. *Epidemiology*. 2010;21(3):383-388. doi:[10.1097/EDE.0b013e3181d61eeb](https://doi.org/10.1097/EDE.0b013e3181d61eeb)
- 185.** Greenland S. Basic methods for sensitivity analysis of biases. *Int J Epidemiol*. 1996;25(6):1107-1116. doi:[10.1093/ije/25.6.1107](https://doi.org/10.1093/ije/25.6.1107)
- 186.** Robins JM, Rotnitzky A, Scharfstein DO. Sensitivity analysis for selection bias and unmeasured confounding in missing data and causal inference models. In: Halloran ME, Berry D, eds. *Statistical Models in Epidemiology, the Environment, and Clinical Trials*. Springer; 2000:1-94. doi:[10.1007/978-1-4612-1284-3_1](https://doi.org/10.1007/978-1-4612-1284-3_1)
- 187.** Lash TL, Fox MP, MacLehose RF, Maldonado G, McCandless LC, Greenland S. Good practices for quantitative bias analysis. *Int J Epidemiol*. 2014;43(6):1969-1985. doi:[10.1093/ije/dyu149](https://doi.org/10.1093/ije/dyu149)
- 188.** VanderWeele TJ, Ding P. Sensitivity analysis in observational research: introducing the E-value. *Ann Intern Med*. 2017;167(4):268-274. doi:[10.7326/M16-2607](https://doi.org/10.7326/M16-2607)
- 189.** Lash TL, Fox MP, Fink AK. *Applying Quantitative Bias Analysis to Epidemiologic Data*. Springer; 2009. doi:[10.1007/978-0-387-87959-8](https://doi.org/10.1007/978-0-387-87959-8)
- 190.** Vanderweele TJ, Arah OA. Bias formulas for sensitivity analysis of unmeasured confounding for general outcomes, treatments, and confounders. *Epidemiology*. 2011;22(1):42-52. doi:[10.1097/EDE.0b013e3181f74493](https://doi.org/10.1097/EDE.0b013e3181f74493)
- 191.** MacLehose RF, Ahern TP, Lash TL, Poole C, Greenland S. The importance of making assumptions in bias analysis. *Epidemiology*. 2021;32(5):617-624. doi:[10.1097/EDE.0000000000001381](https://doi.org/10.1097/EDE.0000000000001381)
- 192.** Toh S, Hernán MA. Causal inference from longitudinal studies with baseline randomization. *Int J Biostat*. 2008;4(1):22. doi:[10.2202/1557-4679.1117](https://doi.org/10.2202/1557-4679.1117)
- 193.** Hernán MA, Robins JM. Per-protocol analyses of pragmatic trials. *N Engl J Med*. 2017;377(14):1391-1398. doi:[10.1056/NEJMsm1605385](https://doi.org/10.1056/NEJMsm1605385)
- 194.** Rosenbaum PR. *Transparency: Design of Observational Studies*. Springer; 2020:175-177. doi:[10.1007/978-3-030-46405-9_6](https://doi.org/10.1007/978-3-030-46405-9_6)
- 195.** Cox DR. Applied statistics: a review. *Ann Appl Stat*. 2007;1(1):1-16. doi:[10.1214/07-AOAS113](https://doi.org/10.1214/07-AOAS113)
- 196.** Greenland S. Invited commentary: the need for cognitive science in methodology. *Am J Epidemiol*. 2017;186(6):639-645. doi:[10.1093/aje/kwx259](https://doi.org/10.1093/aje/kwx259)